

ZAPORIZHZHIA, Ukraine

WMO: 346010

Lat: 47.8786N

Lon: 35.0825E

Elev: 367

StdP: 14.50

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	5.8	-5.6	4.1	1.2	0.3	5.6	7.0	23.0	21.4	21.0	23.7	8.3	350	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	94.0	68.6	91.0	68.0	87.7	66.9	72.0	85.7	70.6	84.2	69.2	82.5	9.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.1	104.5	76.7	66.4	98.4	75.2	64.7	92.9	74.2	35.9	85.7	34.7	84.2	33.6	82.9	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.7	16.7	-5.2	98.4	7.6	3.9	-10.6	101.1	-15.1	103.4	-19.3	105.6	-24.8	108.4		
(5)			-5.7	75.2	7.4	1.8	-11.0	76.5	-15.4	77.6	-19.5	78.7	-24.9	80.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	25.9	28.1	37.4	51.1	62.4	69.4	73.7	72.7	62.4	49.5	38.4	29.8	
	DBStd	18.95	10.29	10.15	7.84	7.30	7.61	6.34	5.74	6.68	7.59	8.30	8.45	9.63	
	HDD50	2933	748	614	393	74	5	0	0	0	4	112	358	625	
	HDD65	6208	1213	1034	854	419	142	28	3	10	136	482	797	1090	
	CDD50	3006	0	0	4	106	388	582	736	705	376	98	11	0	
	CDD65	805	0	0	0	1	61	160	273	250	57	3	0	0	
	CDH74	8392	0	0	0	29	634	1567	2822	2770	543	27	0	0	
	CDH80	3343	0	0	0	5	178	553	1192	1253	158	4	0	0	
Wind	WSAvg	8.1	8.8	9.2	9.5	8.6	7.4	7.3	7.1	7.5	7.5	7.6	8.2	8.9	
Precipitation	PrecAvg	20.0	1.6	1.4	1.6	1.5	1.8	2.5	1.8	1.5	1.7	1.4	1.7	1.7	
	PrecMax	30.5	4.3	3.1	3.0	3.3	4.8	6.2	4.8	6.1	6.1	2.7	4.9	3.8	
	PrecMin	14.8	0.4	0.4	0.4	0.0	0.1	0.9	0.4	0.2	0.1	0.1	0.1	0.2	
	PrecStd	3.6	0.9	0.7	0.7	1.0	1.1	1.3	1.3	1.4	1.4	0.7	1.1	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	51.7	63.7	77.4	89.7	93.0	97.7	99.4	89.9	77.2	62.6	52.1	
		MCWB	44.8	47.3	50.1	58.8	66.0	69.1	69.7	68.7	66.4	60.5	56.0	49.8	
	2%	DB	43.2	46.4	58.0	72.0	84.5	89.5	93.5	95.1	84.4	71.3	57.4	48.2	
		MCWB	41.9	43.9	48.0	55.8	63.5	67.4	69.5	68.0	63.8	58.6	53.6	46.9	
	5%	DB	40.8	42.9	53.5	68.3	80.9	86.1	90.4	91.3	80.3	66.3	53.6	44.9	
		MCWB	39.7	40.4	45.7	54.1	62.6	66.7	69.0	67.3	61.6	57.6	51.1	43.8	
	10%	DB	37.4	39.6	49.1	64.7	77.0	82.8	87.2	87.4	76.0	62.0	50.0	42.4	
		MCWB	36.6	37.9	43.3	52.1	61.4	65.8	68.0	66.6	60.4	55.5	48.2	41.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	48.4	52.4	60.3	69.3	72.8	74.6	73.0	69.2	63.8	57.4	50.6
			MCDB	46.7	50.8	60.7	72.9	81.7	84.7	87.4	88.9	82.9	72.0	60.8	51.9
2%		WB	42.3	44.2	49.3	57.7	66.5	70.8	72.5	71.0	66.6	61.0	54.4	46.8	
		MCDB	43.1	45.9	56.0	68.6	78.7	82.7	85.9	86.1	77.8	67.4	57.2	47.7	
5%		WB	39.4	40.7	46.4	55.6	64.5	69.1	71.1	69.7	64.4	58.6	51.4	43.9	
		MCDB	40.3	42.2	51.3	65.4	76.2	80.8	84.7	84.7	74.4	64.5	53.3	44.8	
10%		WB	36.5	38.2	43.9	53.8	62.7	67.5	69.7	68.1	62.5	56.0	48.2	41.1	
		MCDB	37.2	39.5	48.4	62.7	74.4	78.9	83.1	82.1	72.3	61.4	49.6	42.0	
Mean Daily Temperature Range	5% DB	MDBR	9.6	11.2	14.4	19.1	20.3	19.8	20.9	21.8	19.7	16.0	10.9	9.0	
		MCDBR	9.5	13.6	22.2	25.0	24.9	24.0	24.9	26.2	25.0	21.9	15.4	10.7	
		MCWBR	8.5	11.0	13.8	12.5	11.4	9.5	9.1	9.5	11.1	12.5	11.1	9.6	
	5% WB	MCDBR	9.2	12.3	19.3	21.9	22.2	20.8	21.8	22.8	20.7	19.2	13.2	10.2	
		MCWBR	8.5	10.7	13.3	11.9	11.2	9.8	9.1	9.9	10.6	12.4	10.4	9.5	
Clear-Sky Solar Irradiance	taub	0.332	0.360	0.389	0.449	0.416	0.416	0.440	0.438	0.412	0.374	0.353	0.334		
	taud	2.230	2.184	2.216	2.082	2.270	2.293	2.232	2.226	2.269	2.354	2.324	2.282		
	Ebn at Noon	226	246	259	254	269	269	260	255	250	241	218	210		
	Edh at Noon	25	33	38	48	41	40	42	41	36	28	23	22		
All-Sky Solar Radiation	RadAvg	307	581	948	1459	1877	1994	1987	1789	1250	754	348	228		
	RadStd	55	100	117	132	156	160	112	104	137	106	54	42		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.39	N/A	N/A	+2.27	N/A	N/A	N/A	-263	+209	+131
(47) Regional (9 neighbors)		+1.40	N/A	N/A	+2.05	N/A	N/A	-250	-356	+236	+155

Nomenclature: See separate page